EVENTUAL PROPERNESS OF THE TRANSITION MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Chow's lemma. Cohomology of Schemes, Chow lemmas

Source reference. Original reference 081D: sources/stacks-chow/tags.txt, source lines 10213–10213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZQ: sources/stacks-chow/tags.txt, source lines 2603–2603. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03GI: sources/stacks-chow/tags.txt, source lines 4529–4529. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0202: sources/stacks-chow/tags.txt, source lines 2619–2619. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W3: sources/stacks-chow/tags.txt, source lines 2468–2468. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W4: sources/stacks-chow/tags.txt, source lines 2469–2469. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W6: sources/stacks-chow/tags.txt, source lines 2471–2471. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03GN: sources/stacks-chow/tags.txt, source lines 4534–4534. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZP: sources/stacks-chow/tags.txt, source lines 2602–2602. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Assumptions and notation as in Situation 081D. If

(1) *f is proper, and*
(2) *f_0 is locally of finite type,*

then there exists an i such that f_i is proper.

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Proof. By Lemma 01ZQ we see that f_i is separated for some $i \geq 0$. Replacing 0 by i we may assume that f_0 is separated. Observe that f_0 is quasi-compact, see Schemes, Lemma 03GI. By Lemma 0202 we can choose a diagram

$$\begin{array}{ccccc} X_0 & \xleftarrow{\pi} & X_0' & \longrightarrow & \mathbf{P}^n_{Y_0} \\ & \searrow & \downarrow & \swarrow & \\ & & Y_0 & & \end{array}$$

where $X_0' \to \mathbf{P}^n_{Y_0}$ is an immersion, and $\pi : X_0' \to X_0$ is proper and surjective. Introduce $X' = X_0' \times_{Y_0} Y$ and $X_i' = X_0' \times_{Y_0} Y_i$. By Morphisms, Lemmas 01W3 and 01W4 we see that $X' \to Y$ is proper. Hence $X' \to \mathbf{P}^n_Y$ is a closed immersion (Morphisms, Lemma 01W6). By Morphisms, Lemma 03GN it suffices to prove that $X_i' \to Y_i$ is proper for some i. By Lemma 01ZP we find that $X_i' \to \mathbf{P}^n_{Y_i}$ is a closed immersion for i large enough. Then $X_i' \to Y_i$ is proper and we win. $\square$